SLIC Index V3

Methodology Technical Paper

English master edition

Published snapshot: 2026-04-29T08:23:20.632Z

This paper documents the live published SLIC scorer as implemented in **scripts/rescore-all-cities.mjs** and written to **src/data/publishedRankingData.json**. It replaces earlier draft descriptions of a 35-indicator anchor model that no longer matches the public dataset.

Snapshot fact	Value
Cities in public file	163
Ranked cities	159
Watchlist cities	4
Scored metric lines	20
Visible non-scoring diagnostics	3

1. Published method

SLIC has five public pillars. Within each pillar, observed scored metrics are combined by weighted mean. Across pillars, the final score uses a weighted AMPI, so cross-pillar imbalance is punished mathematically rather than rhetorically.

```
normalize(x, p05, p95, positive) = 100 * clamp((winsor(x) - p05) / (p95 - p05), 0, 1)
normalize(x, p05, p95, negative) = 100 * clamp((p95 - winsor(x)) / (p95 - p05), 0, 1)

P_p(c) = sum observed alpha_(p,m) * q_m(c) / sum observed alpha_(p,m)

mu(c)  = (25 Pressure + 22 Viability + 18 Capability + 15 Community + 20 Creative) / 100
var(c) = (25(Pressure-mu)^2 + 22(Viability-mu)^2 + 18(Capability-mu)^2 +
          15(Community-mu)^2 + 20(Creative-mu)^2) / 100
AMPI(c) = mu(c) - var(c)/mu(c)
SLIC(c) = max(0, AMPI(c) - coverage_penalty(c))
```

The final published score is therefore **not** a simple weighted average of the displayed pillar scores. That distinction matters: a city with one catastrophic pillar is supposed to score worse than a city with the same weighted mean but more even performance.

2. What is scored

The current public model scores **20** metric lines and keeps **3** additional metric lines visible as diagnostics only. Visibility and scoring are intentionally separated.

Growth (25)

Metric	Weight	Status	Method note
Tax-adjusted PPP disposable income	9	Scored	Residual money after essentials, converted once through the PPP private-consumption factor.
Housing burden	5	Scored	Housing cost share of income; higher burden scores worse.
Household debt burden	4	Scored	Debt relative to income, with fallback when city debt evidence is thin.
Working time pressure	4	Scored	Higher work burden scores worse.
Suicide and severe mental strain	3	Scored	Public-health strain proxy; higher severe-strain burden scores worse.
Economic growth momentum	-	Diagnostic only	Visible diagnostic line; not part of the current aggregate.

Viability (22)

Metric	Weight	Status	Method note
Personal safety	5	Scored	Harm and victimization outcomes, not surveillance theatre.
Transit access and commute burden	5	Scored	Better access and lower burden score better.
Clean air	4	Scored	Higher pollution exposure scores worse.
Water, sanitation, and utility reliability	4	Scored	Safer access and higher reliability score better.
Digital infrastructure	4	Scored	Broadband quality and affordability.
Climate and sunlight livability	-	Diagnostic only	Visible diagnostic line; not part of the current aggregate.

Capability (18)

Metric	Weight	Status	Method note
Healthcare quality	8	Scored	Effective-care quality and access.
Education quality	6	Scored	Learning quality and skills formation.

Metric	Weight	Status	Method note
Equal opportunity and distributional fairness	4	Scored	Composite scored metric: 70% equal-opportunity evidence, 30% reversed Gini context.

Community (15)

Metric	Weight	Status	Method note
Hospitality and belonging	5	Scored	Do people feel welcome in practice?
Tolerance and pluralism	5	Scored	Composite scored metric: 40% Equaldex, 30% Freedom House, 30% reversed hate-crime or civil-liberties proxy.
Cultural, historic, and public-life vitality	5	Scored	Everyday public-life texture; visitor demand is contextual, not an automatic bonus.
Birth-rate optimism	-	Diagnostic only	Visible diagnostic line; not part of the current aggregate.

Creative (20)

Metric	Weight	Status	Method note
Entrepreneurial dynamism	6	Scored	Startup formation and productive entry.
Innovation and research intensity	5	Scored	Research depth and knowledge production.
Economic vitality and productive context	5	Scored	Composite scored metric: 50% investment signal, 30% GDP per capita PPP, 20% GDP growth.
Administrative and investment friction	4	Scored	Higher time-cost and licensing burden score worse.

3. Composite metrics

Three scored terms are composites. When one component is missing, the composite is reweighted over the observed component weights only; nothing is silently imputed.

```
EqualOpportunityFairness = 0.7 * EqualOpportunity + 0.3 * ReverseGini
TolerancePluralism      = 0.4 * Equaldex + 0.3 * FreedomHouse + 0.3 * ReverseHateCrime
EconomicVitality        = 0.5 * InvestmentSignal + 0.3 * GDPperCapitaPPP + 0.2 * GDPGrowth
```

4. Coverage, watchlist logic, and ranking protocol

```
cov_direct = 1 if observed else 0
cov_composite = observed component weight / total component weight

Cov_p(c) = sum alpha_(p,m) * cov_m(c) / sum alpha_(p,m)
Cov(c)   = (25 Cov_pressure + 22 Cov_viability + 18 Cov_capability +
            15 Cov_community + 20 Cov_creative) / 100
```

Coverage grade	Rule	Penalty	Rank status
A	Cov >= 0.75	0	Ranked
B	0.50 <= Cov < 0.75	5	Ranked
C	0.35 <= Cov < 0.50	15	Ranked
Watchlist	Cov < 0.35 or manual watchlist override	0	Watchlist

After score computation, ranked cities receive a pure score rank from the exact unrounded score field: descending SLIC, then deterministic tie-breaks on Community, Pressure, and city label. Alpha, Beta, and Gamma are then computed as a separate public overlay. The overlay allows one city per country across all three tiers, except Taiwan may hold 2 public-tier seats and Japan may hold 2 public-tier seats; Alpha requires Community >= 40 and Pressure >= 40, with Europe capped at 2 Alpha seats, Oceania capped at 0, South Korea capped at 1, Japan capped at 1, Israel excluded from Alpha by country, and the editorial city list (Tokyo) also excluded from Alpha — these cities are barred from Alpha because their median-resident cost-of-living puts them outside SLIC's positive showcase; Beta raises the floors to Community >= 45 and Pressure >= 45; Gamma fills from the remaining ranked cities.

The published JSON snapshot also carries a machine-readable publication manifest, change summary, provenance diagnostics, and a small floor-sensitivity stability analysis so the paper, the dataset, and the audit trail stay aligned.

Live example: Singapore is currently pure rank **#15** with SLIC **60.6**, but its Community (38.8) and Pressure (43.3) place it in **Gamma**. Bangkok is pure rank **#27** with SLIC **58.5**, yet enters **Alpha** because Community (90.0) and Pressure (45.4) both clear the Alpha floor.

5. Worked example from the live dataset

Example city: **Kaohsiung**. This walkthrough uses the published row in the live dataset, not a fictional demonstration.

```
Negative-direction normalization example (personal safety)
raw = 0.8
p05 = 0.23
p95 = 22.63
score = 100 * (22.63 - 0.8) / (22.63 - 0.23)
score = 97.5

Growth weighted mean
Growth = (9x81.5 + 5x81.1 + 4x41.4 + 3x27.5) / 21
Growth = 66.1

Cross-pillar AMPI
mu  = 70.307
var = 252.168
AMPI = 70.307 - 252.168 / 70.307
AMPI = 66.720
Coverage = 0.77 -> grade A -> penalty 0
Published SLIC = 66.7
```

The important methodological point is visible in the arithmetic: the weighted pillar mean for this city is higher than the final score because AMPI makes uneven pillar structure costly.

6. Data profile of the current public file

Measure	Value
City / metro metric share	51.1%
National / international metric share	24.8%
Composite metric share	14.9%
Derived metric share	9.1%
Grade A cities	138
Grade B cities	2
Grade C cities	21
Watchlist cities	4

These shares are computed directly from non-null metric lines in the published JSON. They are included here so the paper does not rely on invented methodology graphics or unsupported source-mix claims.

7. Published ranked cities

The table below is the ranked slice of the current published dataset. Watchlist cities are excluded from this list by definition.

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
1	Raleigh	United States	70.0	75.1	81.6	79.1	53.8	62.0	A
2	Montreal	Canada	69.1	62.3	94.9	75.3	74.1	54.1	A
3	Kaohsiung	Taiwan	66.7	66.1	86.5	90.6	47.1	56.9	A
4	Pittsburgh	United States	66.2	76.0	73.3	79.1	50.2	55.7	A
5	Minneapolis	United States	66.0	71.6	76.4	79.1	44.2	62.0	A
6	Eindhoven	Netherlands	64.3	64.6	98.2	71.8	65.7	42.8	A
7	Graz	Austria	64.2	62.4	90.1	74.4	73.4	41.4	A
8	Taipei	Taiwan	63.8	48.4	89.5	89.7	47.1	69.4	A
9	Ottawa	Canada	63.6	55.3	94.9	75.3	75.6	44.2	A
10	Prague	Czechia	63.5	62.0	90.6	61.9	87.7	42.4	A
11	Jeju City	South Korea	61.6	56.2	98.1	85.8	44.8	50.6	A
12	Chicago	United States	60.9	56.0	66.8	79.1	47.6	62.0	A
13	Braga	Portugal	60.7	58.4	96.7	68.8	57.9	43.8	A
14	Lyon	France	60.6	52.4	98.4	63.9	73.1	45.1	A
15	Singapore	Singapore	60.6	43.3	90.4	94.1	38.8	73.3	A
16	Milan	Italy	60.5	88.5	85.8	66.9	58.4	29.4	A
17	Busan	South Korea	60.4	53.2	95.9	85.8	44.8	50.6	A
18	Suwon	South Korea	60.3	53.3	93.1	85.8	45.1	50.6	A
19	Incheon	South Korea	60.3	53.3	93.2	85.8	44.8	50.6	A
20	Perth	Australia	60.0	45.3	92.7	87.1	56.3	51.2	A
21	Copenhagen	Denmark	59.9	33.1	98.4	71.3	89.6	61.5	A
22	Tokyo	Japan	59.5	46.2	94.2	76.8	70.2	44.2	A
23	Toronto	Canada	59.4	38.7	94.9	75.3	72.5	54.1	A
24	Valparaiso	Chile	59.4	58.3	76.4	73.6	56.9	43.0	A
25	Adelaide	Australia	58.7	43.7	92.7	87.1	56.3	49.2	A
26	Porto	Portugal	58.6	52.5	96.7	68.8	57.9	43.8	A
27	Bangkok	Thailand	58.5	45.4	84.9	62.5	90.0	46.0	A
28	Melbourne	Australia	58.5	36.4	92.7	87.1	56.3	60.5	A
29	Brisbane	Australia	58.4	39.7	92.7	87.1	56.3	54.4	A
30	Zurich	Switzerland	58.3	39.1	98.2	72.7	76.0	49.4	A
31	Vienna	Austria	58.3	46.8	90.1	74.4	71.1	41.4	A
32	Helsinki	Finland	58.3	40.9	93.5	81.0	57.5	52.9	A
33	Tallinn	Estonia	58.3	51.0	92.4	58.4	49.0	57.8	A
34	London	United Kingdom	58.2	35.1	82.8	78.4	70.2	59.3	A
35	Santiago	Chile	57.9	50.7	72.9	83.8	56.9	44.6	A
36	New York	United States	57.7	27.1	85.4	82.0	68.5	75.3	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
37	Fukuoka	Japan	57.6	59.0	95.8	67.4	46.6	42.3	A
38	Dunedin	New Zealand	57.6	51.2	94.3	85.7	57.2	37.1	A
39	Christchurch	New Zealand	57.3	45.8	94.3	85.7	57.2	41.9	A
40	Vancouver	Canada	57.1	34.7	94.9	75.3	70.2	54.1	A
41	Venice	Italy	56.8	53.9	84.7	67.8	68.6	34.8	A
42	Wellington	New Zealand	55.9	37.8	94.3	85.7	57.2	48.8	A
43	Hiroshima	Japan	55.8	59.0	95.8	67.4	46.6	37.3	A
44	Sapporo	Japan	55.8	59.0	95.8	67.4	46.6	37.3	A
45	Torun	Poland	55.7	68.4	77.3	65.1	52.0	31.9	A
46	Kobe	Japan	55.5	58.0	95.8	67.4	46.6	37.3	A
47	Katowice	Poland	54.8	65.9	73.7	65.1	52.0	31.9	A
48	Sydney	Australia	54.7	27.8	92.7	87.1	56.3	63.9	A
49	Gdansk	Poland	54.6	60.9	83.4	65.1	52.0	31.9	A
50	Auckland	New Zealand	54.5	32.4	94.3	85.7	57.2	52.9	A
51	Dubai	United Arab Emirates	54.2	69.8	72.4	62.0	29.5	46.2	A
52	Abu Dhabi	United Arab Emirates	54.1	70.5	70.8	62.0	29.5	46.2	A
53	Bologna	Italy	54.0	56.4	87.9	66.9	58.4	29.4	A
54	Bratislava	Slovakia	53.9	57.3	91.2	49.7	63.2	35.3	A
55	Amsterdam	Netherlands	53.9	37.6	98.2	71.8	65.7	42.8	A
56	Manama	Bahrain	53.9	66.4	66.9	57.8	39.7	42.5	A
57	Hobart	Australia	52.6	33.8	92.7	87.1	56.3	45.1	A
58	Krakow	Poland	51.9	53.5	74.6	65.1	52.0	31.9	A
59	Port Louis	Mauritius	51.5	56.0	83.6	47.0	39.6	45.4	A
60	Montevideo	Uruguay	51.1	52.4	72.0	74.6	64.6	24.4	A
61	San Juan	Puerto Rico	49.8	43.8	68.8	68.3	63.3	45.8	B
62	Bucharest	Romania	49.5	57.5	85.8	50.6	40.7	34.8	A
63	Melaka	Malaysia	49.5	63.0	75.3	52.4	40.3	31.5	A
64	Kota Kinabalu	Malaysia	49.3	61.5	77.9	52.4	40.0	31.5	A
65	Kuching	Malaysia	49.1	61.5	72.0	52.4	41.1	31.5	A
66	Paris	France	48.6	27.7	98.4	63.9	69.2	45.1	A
67	George Town	Malaysia	48.4	58.0	71.1	52.4	41.1	31.5	A
68	Budapest	Hungary	48.1	47.2	94.7	55.5	63.5	26.2	A
69	San Jose	Costa Rica	48.0	45.7	55.9	66.0	47.9	35.9	A
70	Panama City	Panama	47.9	61.0	61.2	64.2	41.8	26.9	A
71	Kuala Lumpur	Malaysia	46.8	51.0	72.4	52.4	41.1	31.5	A
72	Hat Yai	Thailand	46.6	55.1	66.7	56.4	52.2	22.8	A
73	Cordoba	Argentina	45.7	67.5	78.7	78.6	46.8	10.2	A
74	Salalah	Oman	45.4	73.9	71.7	42.0	31.2	30.6	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
75	Male	Maldives	45.1	54.6	80.7	39.8	35.8	35.4	A
76	Phuket	Thailand	44.9	48.0	68.0	56.4	52.2	22.8	A
77	Curitiba	Brazil	44.9	51.8	58.0	69.5	36.3	27.6	A
78	Khobar	Saudi Arabia	44.7	91.3	66.7	78.9	17.2	24.1	A
79	Jeddah	Saudi Arabia	44.7	90.6	66.7	78.9	17.2	24.1	A
80	Riyadh	Saudi Arabia	44.7	89.9	66.7	78.9	17.2	24.1	A
81	Florianopolis	Brazil	44.5	50.4	58.0	69.5	36.3	27.6	A
82	Chiang Mai	Thailand	43.8	54.3	49.7	56.4	52.2	22.8	A
83	Muscat	Oman	43.6	66.9	60.3	42.0	31.2	30.6	A
84	Belgrade	Serbia	43.1	46.5	72.0	53.2	32.1	30.3	A
85	Doha	Qatar	42.9	59.0	56.5	54.4	18.6	37.6	A
86	Buenos Aires	Argentina	42.7	55.4	78.7	78.6	46.8	10.2	A
87	Tianjin	China	42.6	43.5	72.6	71.9	19.7	37.1	A
88	Vilnius	Lithuania	42.6	65.0	100.0	41.6	74.0	41.4	C
89	Sao Paulo	Brazil	42.3	42.7	58.0	69.5	36.3	27.6	A
90	Chongqing	China	41.0	38.5	75.2	71.9	20.3	37.1	A
91	Chengdu	China	40.9	37.8	77.0	71.9	20.5	37.1	A
92	Guangzhou	China	40.5	38.1	91.0	71.9	19.7	37.1	A
93	Gothenburg	Sweden	40.3	65.3	100.0	14.5	95.5	63.3	C
94	Shenzhen	China	39.1	35.3	92.7	71.9	19.7	37.1	A
95	Gaborone	Botswana	39.0	51.4	45.3	22.2	45.2	39.0	A
96	Hangzhou	China	39.0	33.5	84.0	71.9	20.6	37.1	A
97	Kuwait City	Kuwait	38.5	91.4	52.9	65.0	19.7	19.9	A
98	Nanjing	China	38.4	52.6	75.3	59.0	7.1	34.5	A
99	Suva	Fiji	38.4	45.8	51.2	47.7	30.8	25.1	A
100	Riga	Latvia	38.3	60.6	100.0	33.5	76.2	40.3	C
101	Arequipa	Peru	38.1	57.5	40.8	42.3	34.9	24.7	A
102	Bogota	Colombia	37.8	48.7	39.2	66.3	41.9	19.6	A
103	Medellin	Colombia	37.7	54.4	35.2	66.3	41.9	19.6	A
104	Shanghai	China	37.6	31.3	85.2	71.9	19.7	37.1	A
105	Lima	Peru	36.7	51.3	37.3	42.3	34.9	24.7	A
106	Windhoek	Namibia	36.7	42.7	51.9	32.1	39.3	25.4	A
107	Nizhny Novgorod	Russia	36.6	70.7	70.8	59.2	15.9	15.5	A
108	Cork	Ireland	34.8	82.1	100.0	13.3	91.8	37.4	C
109	Cebu City	Philippines	34.6	55.8	44.8	40.4	30.8	17.5	A
110	Makati	Philippines	34.4	52.1	45.7	40.4	30.8	17.5	A
111	Valencia	Spain	34.0	59.8	100.0	31.1	87.1	30.4	C
112	Bergen	Norway	32.9	76.4	100.0	3.2	97.8	48.8	C
113	Zagreb	Croatia	32.7	61.9	85.0	17.8	77.4	42.3	C
114	Da Nang	Vietnam	32.6	51.7	77.6	44.3	28.1	20.8	B

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
115	Cape Town	South Africa	32.6	34.2	35.1	26.3	42.3	29.8	A
116	Merida	Mexico	32.4	53.7	29.3	55.5	40.5	14.1	A
117	Rabat	Morocco	32.4	46.9	66.7	34.1	24.2	18.8	A
118	Casablanca	Morocco	32.3	46.0	62.3	34.1	24.2	18.8	A
119	Johannesburg	South Africa	31.7	34.3	31.1	26.3	42.3	29.8	A
120	Surabaya	Indonesia	31.5	63.7	36.2	35.1	29.9	17.8	A
121	Hyderabad	India	31.3	44.7	52.7	30.4	33.5	14.6	A
122	Bengaluru	India	31.2	43.9	54.5	30.4	33.5	14.6	A
123	Jakarta	Indonesia	31.1	58.6	34.5	35.1	29.9	17.8	A
124	Guadalajara	Mexico	30.6	49.9	25.8	55.5	40.5	14.1	A
125	Pune	India	30.6	44.4	44.8	30.4	33.5	14.6	A
126	Chandigarh	India	30.6	43.9	44.1	30.4	34.1	14.6	A
127	Moscow	Russia	30.6	41.0	71.2	59.2	15.9	15.5	A
128	Kigali	Rwanda	30.5	56.0	47.3	15.8	30.5	24.8	A
129	Tbilisi	Georgia	30.5	50.0	77.6	33.1	40.9	43.1	C
130	Kandy	Sri Lanka	30.2	51.5	49.4	33.7	28.6	11.7	A
131	Colombo	Sri Lanka	29.9	48.8	48.5	33.7	28.6	11.7	A
132	Mexico City	Mexico	29.3	39.2	27.1	55.5	40.5	14.1	A
133	Aqaba	Jordan	29.0	43.2	62.5	46.5	11.5	17.0	A
134	Thimphu	Bhutan	28.5	47.2	46.2	25.2	40.4	10.2	A
135	Amman	Jordan	28.3	40.3	59.0	46.5	11.5	17.0	A
136	Antwerp	Belgium	27.0	65.1	100.0	4.4	87.9	40.4	C
137	Santo Domingo	Dominican Republic	26.8	36.9	88.8	53.7	46.6	28.2	C
138	Brno	Czechia	26.2	70.5	92.5	0.0	84.3	42.4	C
139	Nairobi	Kenya	24.3	48.5	53.9	12.5	28.1	14.5	A
140	Asuncion	Paraguay	24.2	40.4	96.3	74.7	45.4	14.9	C
141	Accra	Ghana	23.5	55.6	29.0	14.9	30.3	17.8	A
142	Kumasi	Ghana	22.6	59.4	27.5	14.9	30.3	17.8	A
143	Pokhara	Nepal	22.2	44.3	36.5	12.2	21.7	16.6	A
144	Ljubljana	Slovenia	22.0	67.0	94.4	0.0	90.6	33.2	C
145	Kathmandu	Nepal	21.2	42.3	30.3	12.2	21.7	16.6	A
146	Dhaka	Bangladesh	20.6	49.3	36.9	10.8	19.6	15.7	A
147	Chattogram	Bangladesh	20.5	51.3	36.9	10.8	19.6	15.7	A
148	Phnom Penh	Cambodia	20.1	43.9	55.1	18.2	34.5	36.3	C
149	Dakar	Senegal	18.6	40.5	12.6	10.6	26.3	26.7	A
150	Munich	Germany	13.5	18.3	100.0	27.1	65.0	36.2	C
151	Dar es Salaam	Tanzania	13.5	31.8	70.1	59.8	24.1	12.7	C
152	Islamabad	Pakistan	9.0	47.5	25.0	8.1	13.0	4.4	A
153	Karachi	Pakistan	8.9	48.0	25.0	8.1	13.0	4.4	A
154	Lahore	Pakistan	8.9	48.3	25.0	8.1	13.0	4.4	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
155	Port Vila	Vanuatu	8.7	22.8	100.0	26.7	55.4	21.9	C
156	Apia	Samoa	7.6	27.6	100.0	-	55.5	10.2	C
157	Port Moresby	Papua New Guinea	0.0	18.9	100.0	-	28.6	3.7	C
158	Kampala	Uganda	0.0	31.8	0.0	68.7	15.1	22.5	C
159	Alexandria	Egypt	0.0	45.0	0.0	11.3	7.7	30.6	C